

UX Design of Digital Marketing Management System for Micro, Small, and Medium Enterprises to Increase Culinary Potential of Malang City

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Abstract— The development of digital technology has changed the paradigm of life and fostering new ways of working in the community. Economic activities have shifted to digital form with the presence of social media and marketplaces. Based on data quoted from We are Social, it is known that as many as 170 million people (61.8% of the total population in Indonesia) are active social media users in Indonesia. Thus, the use of social media as a digital branding platform is very attractive and promising. In addition, the era of digital technology disruption makes digital branding and marketing skills very important for digital business people. Thus, digital branding learning needs to be encouraged to increase the competitiveness of the digital era. However, the absence of a platform that has special features for learning digital branding management is a problem. Therefore, as an effort to create a virtual branding application, UI/UX design efforts are needed using a design thinking approach.

Keywords— Digital branding, Social media, Market place, Design thinking.

I. INTRODUCTION

Technological developments in human life today have reached the stage of massive digitalization. Many sectors are shifting from conventional into digital processes in order to increase effectiveness. With these changes, people's behavior slowly began to change according to the movement of technological trends [1]. Changes in behavior in society ultimately form a new paradigm in problem solving. Thus, people's behavior is formed that has a new pattern of working that is more adaptive to the current situation. One of them is people's behavior in the economic field which began to change since a decade ago. In the past, in general, economic processes were carried out in conventional ways. An example is the buying and selling process which requires someone to come directly to a store or market to buy a product. In addition, with technological developments that continue to run, in the end the economic sector also undergoes changes in terms of the process.

The process of economic activity in Indonesia has been transformed into a more modern form. One of them is the shift of buying and selling transactions offline to certain platforms

with the widespread use of social media and marketplace. This is supported by the report data compiled by Hootsuite (We are Social). The report states that as many as 170 million (61.8% of the total population in Indonesia) are active social media users in Indonesia [2]. With this data, the use of social media and marketplaces for the efficiency of the economic sector is very supportive.

The large number of social media users and marketplaces makes digital business players compete to attract consumer interest through various ways. One way that is commonly done is digital branding. Digital branding is the effort of a person or a brand in creating a persona in order to expand the marketing reach of a product or store [3]. Skills are Digital branding digital very important for business people to increase interactions to purchases from consumers. Without an attractive digital branding, a brand will find it difficult to compete with brands other similar.

In addition to the difficulty of competition between brands, one of the reasons why the owner of a brand must have skills digital branding is the bias between the boundaries between the old system and the new system which is currently known as the era of disruption. In the era of disruption, many new innovations have emerged whose presence is not felt and can disrupt the order of the old system and can even abort the system. Amazon became one of the products with disruptive innovation in its time. In 1994, Amazon was present as one of the Start-up that sold books via online. This new innovation was a little difficult to accept at that time, where conventional marketing was still the only option in selling products or services [4].

With the emergence of disruptive innovations, increasing skills digital branding needs to be encouraged for digital business players to be able to compete. The publication of the Central Statistics Agency on E-commerce in 2020, the data shows that 90.18% of business people do their business through digital platforms [5]. The large number of digital business actors is certainly directly proportional to the amount of competition in the digital business industry [6]. However, until now there is no platform that has special features to

improve skills digital branding. Therefore, it is necessary to develop a platform that can provide the freedom to home management skills digital branding.

As an effort to provide a platform that can meet the needs of digital business players, it is necessary to develop a virtual application digital branding. The design of this application must of course be in accordance with the needs of the user. Thus, application developers use a method that can be done in a short time and results with a high level of accuracy [7]. The method used is design thinking. This method is commonly used to develop a software. With a process that is not too complicated, data collection from potential users can be done quickly and with the right target.

In the contemporary digital age, the significance of UI (User Interface) and UX (User Experience) in application design cannot be overstated [8]. The UI, as the interactive bridge between the user and the application, determines the direct accessibility and usability of functionalities, directly influencing user satisfaction. On the other hand, UX focuses on the holistic experience of users, encapsulating not just the direct interactions, but also the emotions, challenges, and gratifications associated with the application usage. A well-designed UI/UX can make the platform more intuitive and easy to use [9]. Given the saturated application market, achieving a seamless integration of both UI and UX has become paramount, acting as a differential advantage. Organizations that underestimate this synergy often find their applications relegated to oblivion, while those that prioritize it witness increased user retention and positive word-of-mouth, underlining the strategic importance of UI/UX in the success trajectory of digital applications.

Given the centrality of UI/UX in determining the success of digital applications, a pressing challenge surfaces when addressing niche domains, such as culinary recognition. Culinary experiences are deeply personal, often intertwined with cultural nuances, individual preferences, and sensory perceptions [10]. Consequently, designing an application for culinary recognition introduces a unique set of challenges. Not only must the application be intuitive and user-friendly in its interface, but it must also resonate with the rich tapestry of culinary experiences, histories, and expectations of its target audience. Thus, the critical problem addressed in this article is the endeavor to conceptualize and implement a UI/UX design for a culinary recognition application that not only meets technical and design standards but is also culturally attuned and universally appealing to its prospective users.

II. LITERATURE REVIEW

A. Digital branding

Digital branding is an effort to design a product or brand using certain digital platforms, such as applications, social media, websites, and so on [11]. Digital branding is one of the most efficient ways to build a persona brand to attract consumers. With digital branding, the presence of a digital brand or business will be easier to recognize. Digital branding is certainly very helpful in shaping marketing strategies. The following is an example of a strategy digital branding.

1) Logo

Logo is a depiction of an entity to the general public. For this reason, a logo must represent the identity of a company or product it represents. So, people will be familiar and quickly recognize a brand through the logo [12].

2) Website

website or site is a collection of information pages that display data in the form of still or motion pictures, text data, animation data, video, audio, and or a combination of all of them. Websites can be static or dynamic. All of these data can form a series of interrelated systems, each of which is linked by page networks (hyperlinks) [13].

A website is said to be static if the information content on the site is only one-way and the information content is directed only to the website operator. It is said to be dynamic when the information content of a website is constantly changing and the information content is interactive with website owners and users.

3) Search Engine Optimization (SEO)

SEO is an effort to optimize a website so that it can enter the top ranking in search results, especially in organic search [14]. Which, organic search is a search made by users or when entering keyword in the browser.

B. Online Learning Platform

Online learning platform is a platform that is generally used to learn something or improve skills in certain things [15]. Simply put, this platform allows the learning process in formal or informal education institutions to take place as usual. Through platforms like this, teachers can still carry out teaching and learning activities remotely. Teachers can interact virtually with their students, teach inmode whiteboard, administer exams, record exam results, and even give grades using report cards. [16]

Almost all activities that usually take place in schools can be done online through this platform. In general, this platform is suitable for distance learning model where students study at home and teachers teach in schools [17].

C. Virtual Laboratory

Virtual laboratory is a set of lab tools in the form of interactive multimedia-based software. Virtual Labs or Virtual Labs, which are operated by computers to simulate lab activities as if the user were in a real lab. [18].

D. Design Thinking

Design thinking is the process by which developers discover problems and find solutions. This requires a high level of empathy and understanding towards the end user, and an iterative process of developing new ideas, making assumptions, to defining problems, with the aim of identifying alternative solutions that may be unusual.

Design thinking is a product's effort to understand user needs more deeply, and create ideas based on data and input from users. After that, the developer tries to make a prototype to be tested again on the user. There are 5 stages in design thinking, namely: empathize, define, invent, prototype, and test. [10].

III. METHOD

The research and development method of user experience design used in achieving this goal is Design thinking with approach User Centered Design (UCD). [19] The application of the method Design thinking with the UCD approach is a product design method that is oriented to user problems and needs. Broadly speaking, this method is grouped into 3 main stages, namely pre-development, development and evaluation,

all of which consist of five sub-stages of design thinking, namely empathize, define, ideation, prototype, and testing. [20].

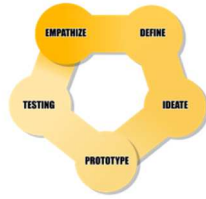


Fig. 1. Five sub-stages of design thinking.

A. Empathize

This stage is a stage to explore the problem to be solved through an empathetic approach. Through this stage, the developer collects data, finds out and validates the problems experienced by users. In addition, developers are also directed to find out what kind of achievements users want,

B. Define

This stage is the stage where developers analyze and synthesize information that has been obtained through the stage empathize. The problem will be further identified to determine the core problem. Thus, developers will find it easier to find troubleshooting solutions.

C. Ideate

This stage is a stage that requires innovation. Where the developer will generate ideas which will then be accommodated and tested to establish ideas that have up-to-date solutions in problem solving. In addition, developers are also expected to be able to provide alternatives if in the future there are development constraints.

D. Prototype

This stage is the stage of making a system based on special features or products in small and inexpensive versions to find out whether the solutions offered have answered the user's problems or testing Prototype can be done by the developer himself, or some lay users.

E. Testing

This stage is the stage of testing and evaluating old products by experts. The results of the evaluation will then be used as a reference for system improvement to refinement in order to obtain optimal results.

The prototype being developed by the E-VLAB website developer has reached the testing stage. So in this paper, the discussion will focus on developing the validity test instrument of this E-VLAB website using the Short User Experience Questionnaire (UEQ) framework based on six scales, namely Attractive, Perpicuity, Efficiency, Dependability, Stimulation, and Novelty. The scale is translated into 8 questions based on the UEQ-S framework with 2 groupings, namely data with Attractiveness, Pragmatic Quality and Hedonic Quality [18].

IV. RESULT AND DISCUSSION

A. Need Assessment

E-VLAB Website Website developed according to user needs to improve management skills digital marketing. As is well known, digital marketing is one of the skills that should be possessed by people who are in the digital industry. This is supported by Ryan's (2016) opinion that digital marketing is

becoming increasingly complex, dynamic, and globalized [19]. Learning about digital marketing is already available on various platforms on the internet. However, until now there is no platform that provides special features for users to manage their digital branding. So, often the digital marketing that is done is still not organized. Even if it is organized, there are difficulties in managing the content digital marketing.

TABLE I. ASSESSMENT CRITERIA

Negative	Positive	Classified
annoying	enjoyable	Attractiveness
bad	good	
unlikeable	pleasing	
unpleasant	pleasant	
unattractive	attractive	
unfriendly	friendly	
slow	fast	Pragmatic Quality
inefficient	efficient	
cluttered	organized	
not understandable	understandable	
difficult to	easy to learn	
complicated	easy	
confusing	clear	
impractical	practical	
unpredictable	predictable	
obstructive	supportive	
not secure	secure	
does not meet expectations	meets expectations	
inferior	valuable	Hedonic Quality
boring	exciting	
not interesting	interesting	
demotivating	motivating	
dull	creative	
conventional	inventive	
usual	cutting edge	
unfriendly	friendly	

In managing content digital marketing, a user-friendly and efficient platform is needed. To achieve this, a development method that refers to user problems is used. So that the development of the E-VLAB platform is carried out using design thinking. With design thinking, the development of a product refers to the needs of potential users. Prospective users of E-VLAB are directed to express their problems in doing digital marketing. Until the problems that are commonly faced by digital business players are found, and through E-VLAB these problems are given solutions.

By using E-VLAB, users are expected to gain skills through management features digital branding that can help with content planning. So, the improvement of a brand digital can be done optimally.

B. Design and Development.



Fig. 2. Landing page of E-VLAB.

This page is the start page of the E-VLAB website. On this page there is a selection of features in the navigation bar section. One of them is a button to login, or register. In addition, there is a search form for users who just want to search for a digital business without registering or logging in.

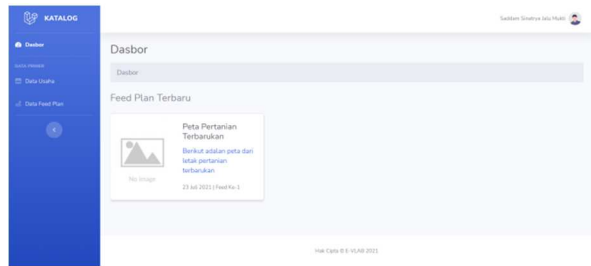


Fig. 3. Dashboard and content page

The dashboard menu contains content plans that will be used for digital branding. If part of the information is opened, it will switch to the Data Feed Plan page which contains details of the schedule for uploading content branding.

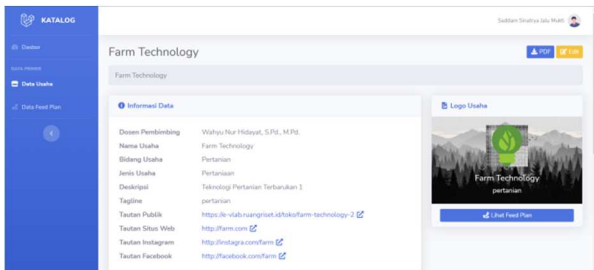


Fig. 4. Business data

This menu contains data from businesses owned by users. On this page, all business data will be stored. Starting from the name to the link of the business.

C. Evaluation

Individual trials conducted by users to assess the validity of the media. Based on the test results, the results obtained are in accordance with Table II which means that E-VLab is a valid media and can be used for learning.

1) Results User Experience Questionnaire (UEQ)

Testing website E-VLAB use UEQ framework generates data that is processed by using a data analysis tool. The data is calculated using a certain formula to get a value with predetermined parameters. The following are the results of data processing based on the UEQ framework.

TABLE II. UEQ TEST RESULT

Short UEQ Scale	
Attractiveness	1.50
Pragmatic Quality	1.30
Hedonic Quality	1.10

Based on the criteria set out in Table I, 8 validation questions measured using the UEQ framework are grouped into three data groups, namely data based on attractiveness, pragmatic quality, and hedonic quality. Where, pragmatic quality represents how the quality of user interaction when doing a job to achieve the desired goal. While the hedonic quality represents user satisfaction in using the website without being related to how the system works in achieving goals.

The data in TABLE III is the final result of data processing that has been collected from respondents using the UEQ framework. Based on the assessment range from UEQ, where the range of positive ratings is in the range > 0.8 while negative ratings are in the range < -0.8 . In the table, it can be seen that the attractiveness of the E-VLAB website has received a positive assessment from respondents with a final score of 1.50. The pragmatic quality of this website has also received a positive assessment from respondents with a final score of 1.30. The last is the hedonic quality which has a final score of 1.10. So overall, through objective data, the E-VLAB website can be categorized as a good website and fulfills user needs.

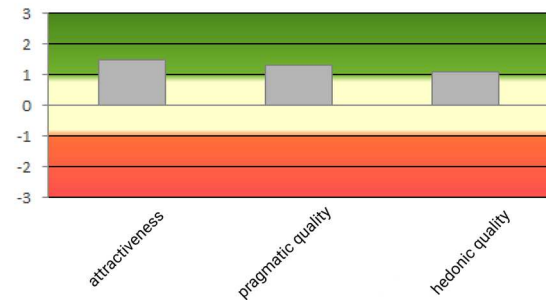


Fig. 5. Statistics

More clearly, the results of data processing are depicted through statistical graphs in figure 5. It can be seen in the graph above that the statistical values of the three categories are in the green zone which indicates a positive assessment.

2) Test Results Usability Testing

E-VLAB website testing is based on aspects that become a benchmark for the efficiency of using a software. In this test, the data taken is subjective data from respondents. The results of this test lead to user satisfaction with the usability of the website. The following are some of the opinions expressed by the respondents.

F-Based on the subjective test results from respondents, there are several things that must be done considered in the development of the E-VLAB website. From the opinions expressed by Respondents 2, and 6 as well as several respondents who have similar opinions, they want integration between the website and social media or Google accounts so content digital marketing that the planned can be uploaded immediately. If the developer can consider this opinion, then

the usability side of the E-VLAB website can develop for the better and answer the needs of users.

TABLE III. USABILITY TEST RESULT

Respondents	Opinion
1	Registration can be made easier by directly using google/ instagram/ other accounts. not filling in the manual
2	Features that allow direct uploads to Social Media
3	It's good, maybe we need to add some help features for users who are new to the application or have just opened a business
4	website can be used properly. Can be added integration with maps to make it easier to see the location
5	website is quite easy to understand. However, having instructions on the use of E-VLAB would be better.
6	It is appropriate. Suggestions, features that are expected if possible can be added reminders that integrate with a google account so that the existing feed plan can be run according to the planned time.
7	Website is good enough. But, because this is for learning, provide examples of good content creation according to business. Such as: - good copy writing - color selection according to business and market - #hashtag selection - productive upload time on every social media
8	It would be better if there was a feature to create a logo

On the other hand, there were responses expressed by several respondents, namely the need to make user manuals or provide modules to make it easier for users who are not too familiar with online management systems through websites. The existence of instructions will certainly be very helpful for users in exploring the features that have been provided by the website [20].

Another opinion that is quite widely expressed is the respondent's desire for this website to have features that can provide examples to steps digital branding good for users. So, in addition to users being able to manage their digital marketing, users also get hands-on learning that can improve their digital marketing skills.

From several opinions that have been expressed by respondents, it can be concluded that in terms of usability, the E-VLAB website is enough to help digital business people in managing their digital marketing. However, there are still some features that can be considered for developers so that they can be developed in the future. A website design approach that makes user needs a reference for development can be realized through the results of this subjective test. It is hoped that developers can realize these features as a form of system development by responding to user needs.

V. CONCLUSIONS

Website E-VLAB digital marketing management system is intended as a learning platform digital management marketing for SMEs culinary student in Malang, this website has many features of which is a feature Dashboard, Feed Plan, Business Data, and so on. The existence of the E-VLAB website is expected to help digital business players, especially students in the city of Malang in order to improve digital marketing skills through digital marketing management. To achieve this goal, it is necessary to design a use experience so

that the platform developed can meet the needs of users appropriately. System testing is carried out using the User Experience Questionnaire (UEQ) framework and Usability Testing. Through the testing process on respondents, the final result is that the E-VLAB website is valid as a whole with a score of 1.50 in terms of attractiveness, a score of 1.30 in terms of pragmatic quality, and a score of 1.10 in terms of hedonic quality based on the UEQ framework, and based on Usability Testing, in general users are satisfied with this website, but there are still some developments that according to users can be developed in order to make it easier for users to manage digital marketing content.

Thus, the development of the E-VLAB website will continue to be carried out in order to optimize the potential for a systemically valid user experience. In the future, by referring to the results of the tests that have been carried out, it is hoped that developers can apply important aspects that can be developed so that the usefulness of the E-VLAB website can be implemented as a more professional digital marketing management system. Thus, the high culinary potential of Malang City can be maximized with digital marketing by digital MSME actors.

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